DHRUV SAINI

dhruvs.pages.dev @ dhruv9saini@gmail.com

EXPERIENCE

Arcanum (Arx Labs)

Head of Physical Engineering | Summer 2026 - Present

- Developing engineering simulations in which AI models design, test, and learn from real-world electrical tasks
- Designing safeguards against **reward hacking** - exploiting scoring loopholes rather than completing engineering tasks
- Promoted from intern; **recruited 2 engineers** and now **managing** their design and validation work

A Sustainable Future

CEO | Oct 2023 - Present

- Built **PCM**, a PySR paper-use predictor reaching **30,000 students** across 150+ schools; secured **$10k** in grants
- Leading **a team of 20** improving school recommendations, saving 1.7M sheets of paper per year

New Energy Technology / AEM Movement

Energy Manager | Summer 2026

- Helped Washington businesses benchmark and reduce building energy use, saving over **600 tons of CO_2 per year**

NASA Glenn High School Engineering Institute | Selected Participant

- Designed and tested jet-noise, spacecraft-power, and lunar-rover prototypes

West Virginia University: SG-REAL Lab

GIS/Software Research Intern | Jul 2025 - Present

- Developing 3D flood-mapping tools to identify vulnerable power-grid nodes and prioritize protection investments

Model Evaluation & Threat Research (METR)

Research Intern | Spring 2025

- Measured how long AI reliably handles multi-step software, ML, cybersecurity, and research tasks (METR Time Horizons)

Dimension

Software Engineer Intern | Jun 2025 - Sept 2025

- Built Stripe, Vercel, GitHub, and other bot integrations; refactored the UI to reduce load time by **70%**

University of Washington

Quantum Technical Writing Intern | Aug 2024 - Sept 2024

- Built Python/React tools to compare revisions of Prof. Anantram's *Quantum Mechanics for Scientists and Engineers*, speeding edits **15x**

PROJECTS

CAD-bench: Benchmarking Language Models on Functional CAD Generation

- Built a benchmark testing whether AI-generated mechanical parts actually work, not just look correct
- Ran physics simulations of gearboxes to uncover mechanical failures that visual AI evaluations miss
- **Accepted to the ICML AIWILD workshop**; code available here and paper available here

EDA Bench: An Execution-Based Benchmark for Language Model Agents on Functional PCB Construction

- Built a benchmark exposing AI-designed circuit boards that look valid but fail in real circuits
- Combined KiCad, circuit simulation, and real-world I/O tests to catch electrical faults before manufacturing
- **Accepted to the ICML FAGEN workshop**; code available here and paper available here

ThinkSwitch: Iterative LoRA training and Weight Interpolation for Low-Compute Improvement on Specific-Purpose Reasoning Tasks

- Used two low-cost methods (small trainable additions and blended model weights) to improve Qwen3-4B on AIME with only $3 in compute
- **Awarded 1st place at the Washington State Science and Engineering Fair**, preprint available here

AWARDS AND HONORS

- **National Science Bowl** (Team Captain) - Founded school's first team; **won Nationwide Virtuals** and **qualified for Nationals**
- **USA AI Olympiad (USAAIO)** - National Finalist, **Bronze Medal**; tested on AI theory, programming, and machine learning
- **American Rocketry Challenge** (Team Captain) - Led five-person team; **qualified for Nationals twice**
- **Earth Prize North America Winner** and **Shane McConkey EcoChallenge 1st Place** for A Sustainable Future
- **National Merit Semifinalist**; **AP Scholar with Distinction**

EDUCATION

Bellevue High School

- Class of 2027; **36 ACT**; 19+ AP courses; 100+ community-service hours; DECA ICDC Top-20 finalist
- President of Programming Club; coaching at Tyee and Odle middle schools, mentoring **200+ students**
- Running Start at Bellevue College; CS 211 (Computer Science II), CS 212 (C++ Data Structures), SEC 450 (Cybersecurity)